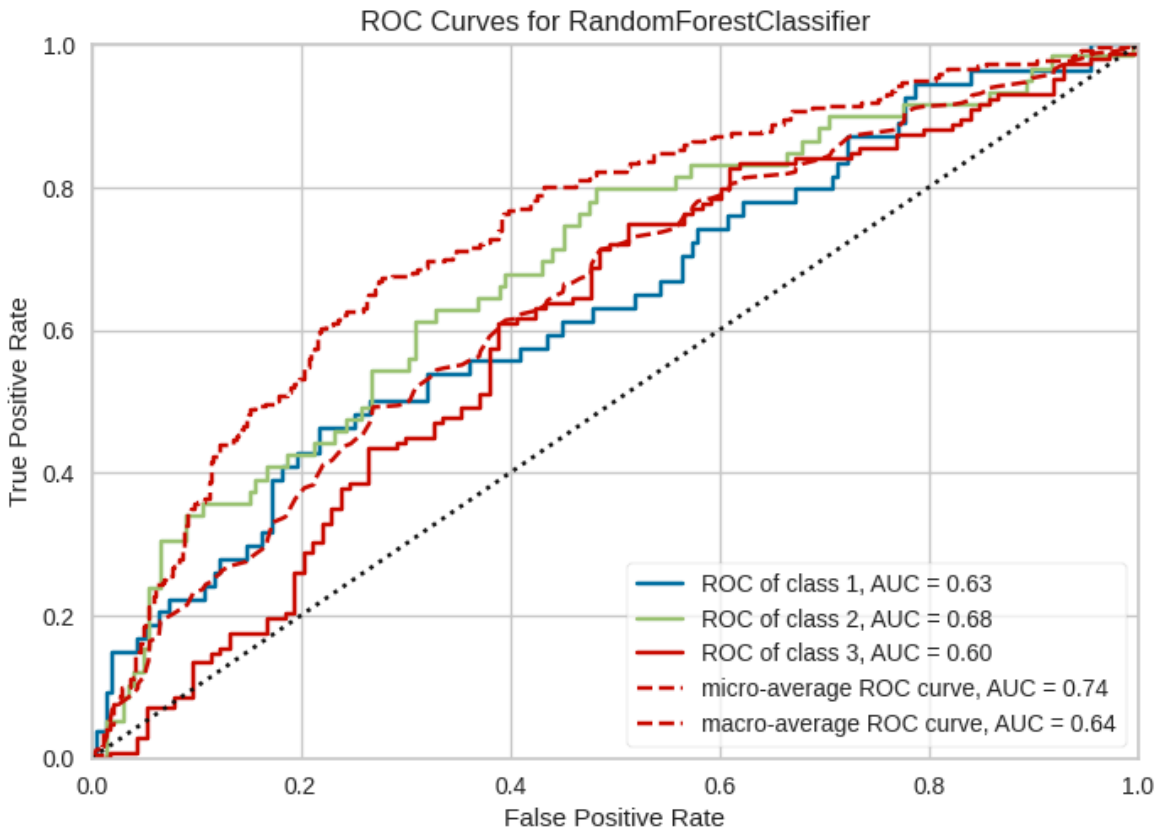
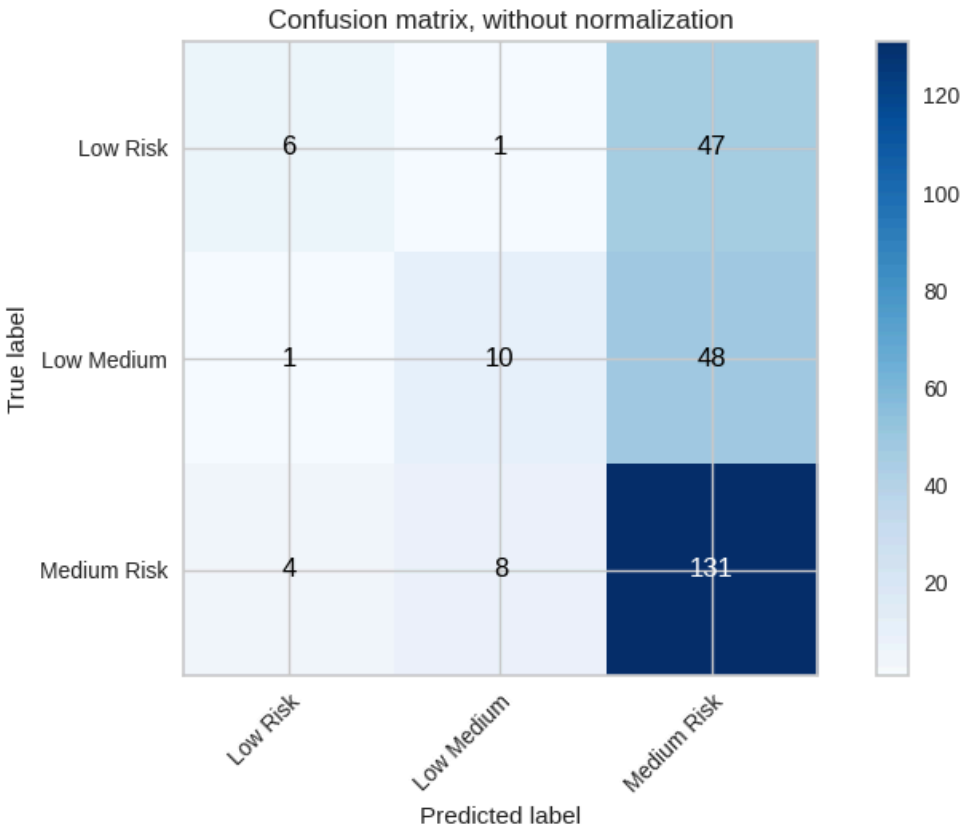
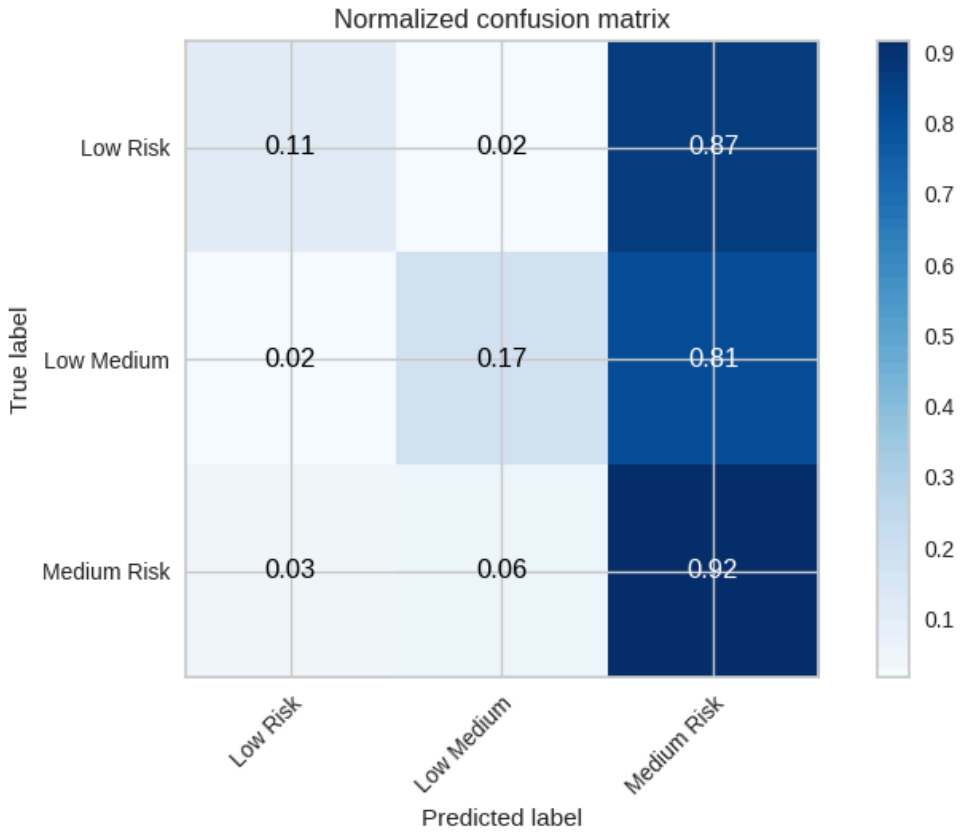
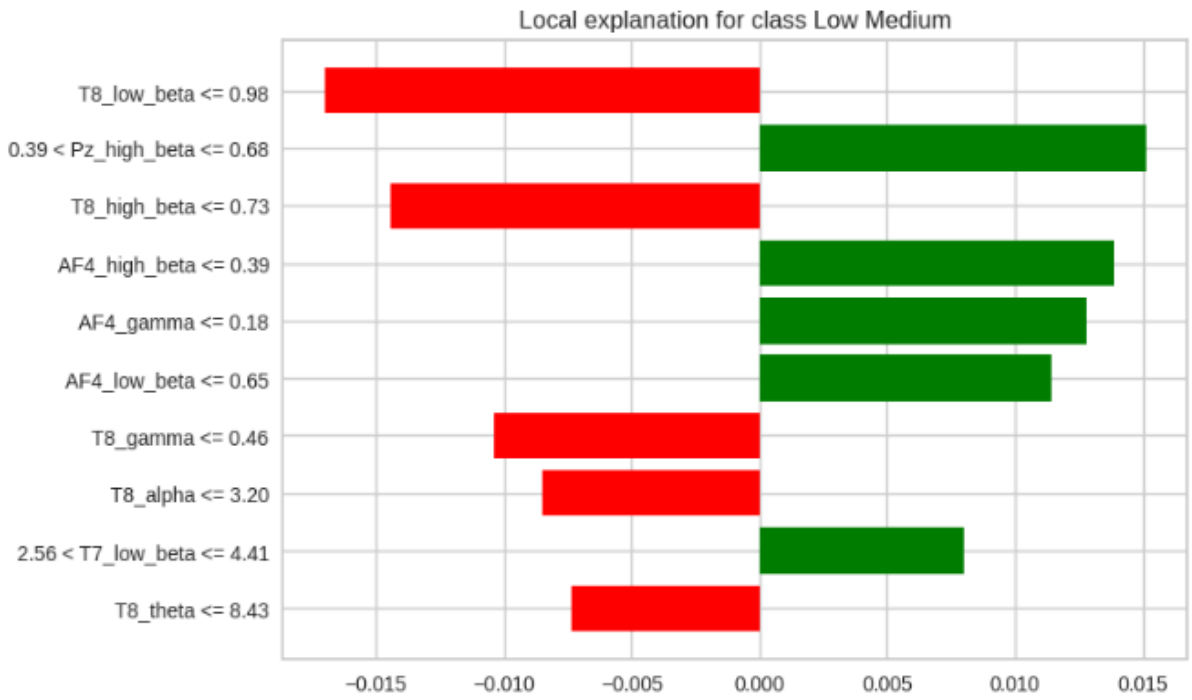
ML Model	Poison rate [%]	Accuracy [%]	Recall [%]	Precision [%]	F1-score [%]
Random Forest	0	57.42	40	55	38
	5	57.03	39	55	36
	25	53.51	36	46	33
	50	48.82	39	41	39
	75	39.45	37	37	36

Poisoning rate 0% :

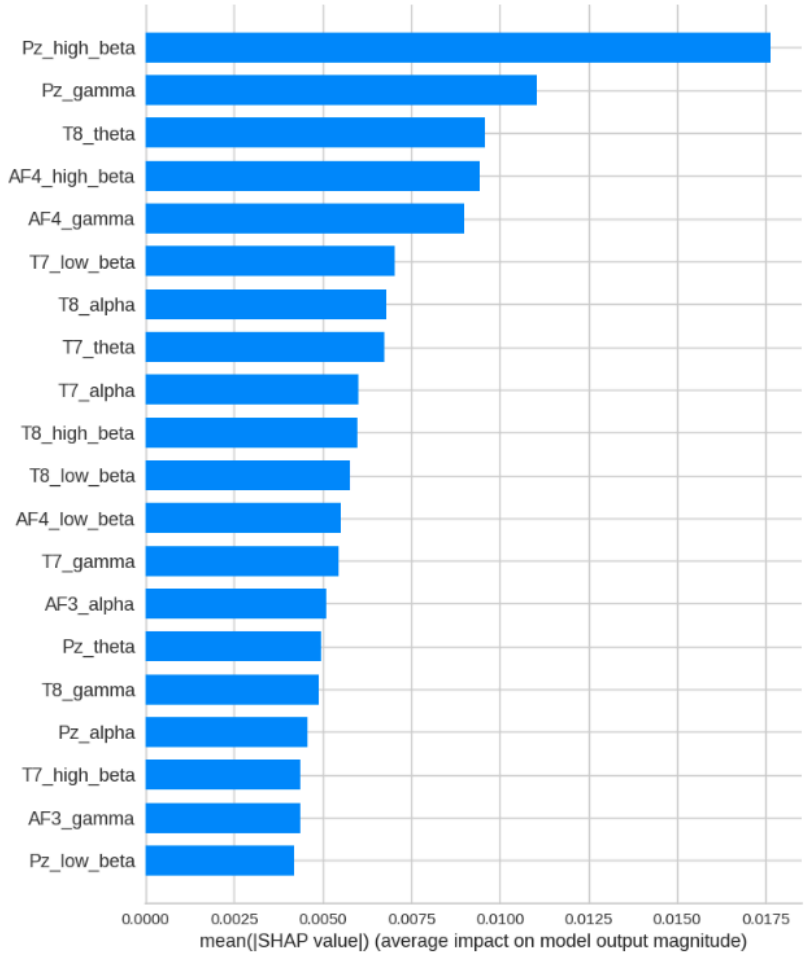


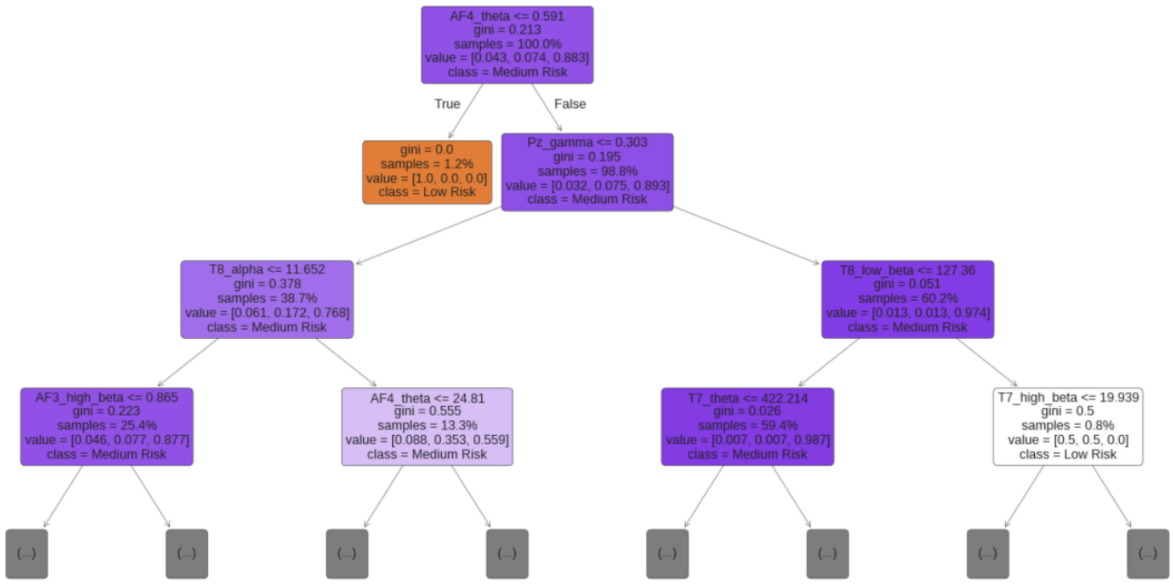

Confusion matrix, without normalization
Low Risk
Low Medium
Medium Risk
True label
6
1
47
1
10
48
4
8
131
Low Risk
Low Medium
Medium Risk
Predicted label
120
100
80
60
40
20


Normalized confusion matrix
Low Risk
Low Medium
Medium Risk
True label
0.11
0.02
0.87
0.02
0.17
0.81
0.03
0.06
0.92
Low Risk
Low Medium
Medium Risk
Predicted label
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1

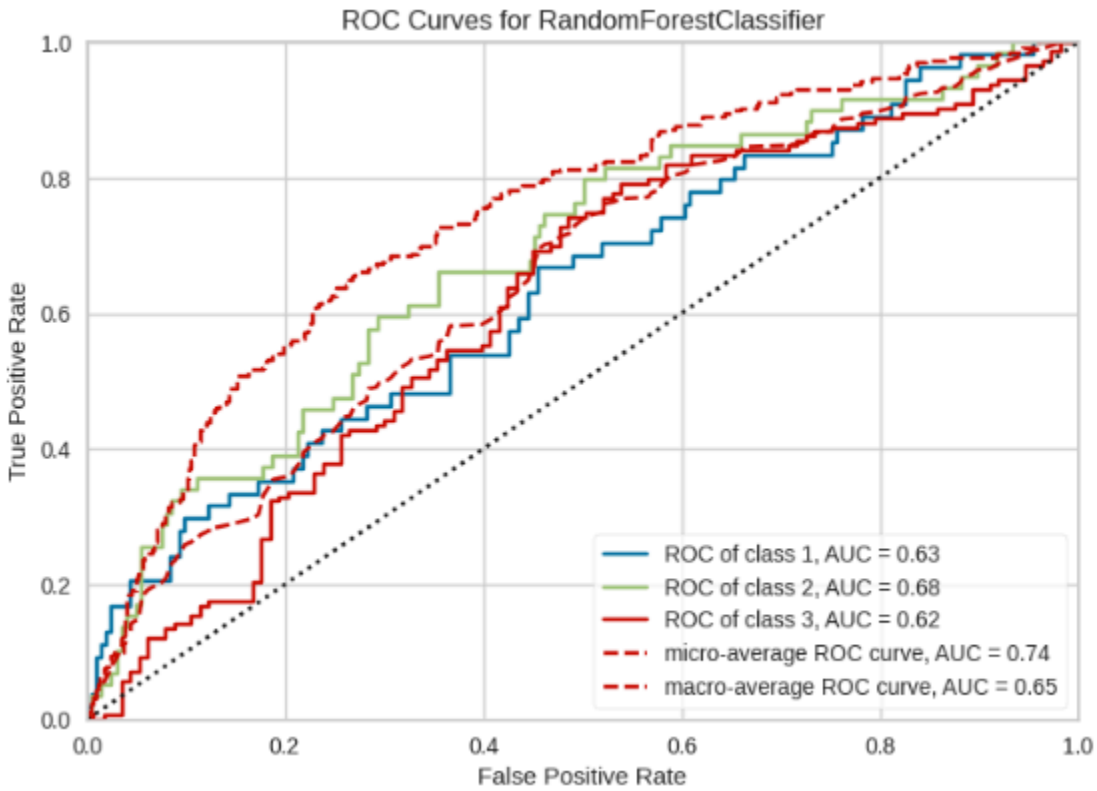


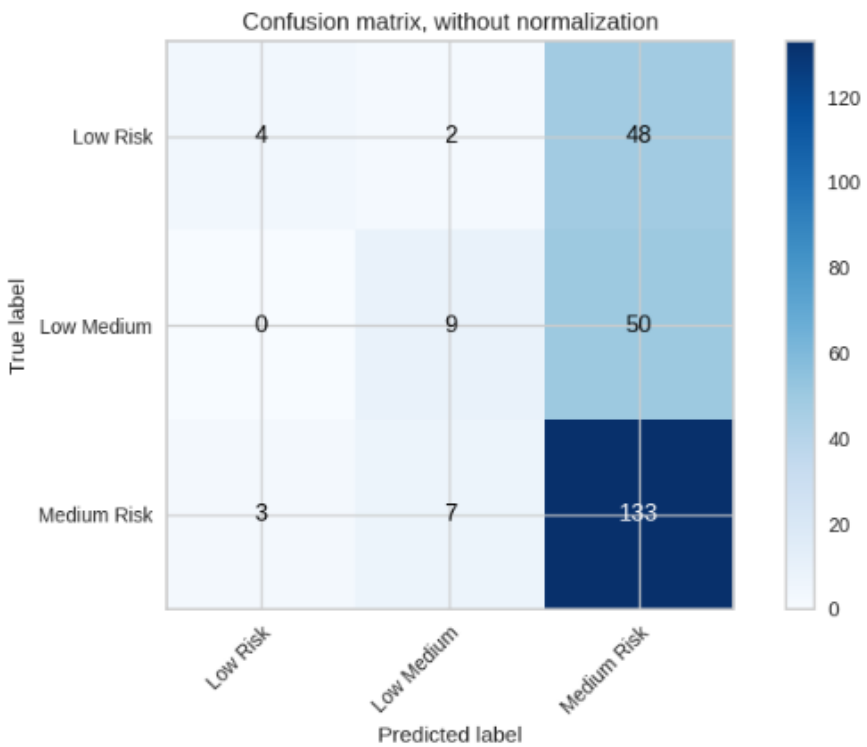
Weight	Feature
0.0242 ± 0.0091	T8_theta
0.0172 ± 0.0161	AF4_gamma
0.0141 ± 0.0094	T7_high_beta
0.0109 ± 0.0151	T8_alpha
0.0094 ± 0.0063	Pz_alpha
0.0086 ± 0.0115	T7_theta
0.0078 ± 0.0070	T7_low_beta
0.0070 ± 0.0115	AF4_alpha
0.0070 ± 0.0091	Pz_gamma
0.0070 ± 0.0058	AF3_gamma
0.0070 ± 0.0104	T8_gamma
0.0070 ± 0.0058	AF3_high_beta
0.0063 ± 0.0106	T7_gamma
0.0055 ± 0.0175	AF4_high_beta
0.0047 ± 0.0091	Pz_high_beta
0.0047 ± 0.0104	Pz_theta
0.0031 ± 0.0031	T8_low_beta
0.0031 ± 0.0077	Pz_low_beta
0.0023 ± 0.0038	AF4_theta
0.0023 ± 0.0106	AF4_low_beta
... 4 more ...	

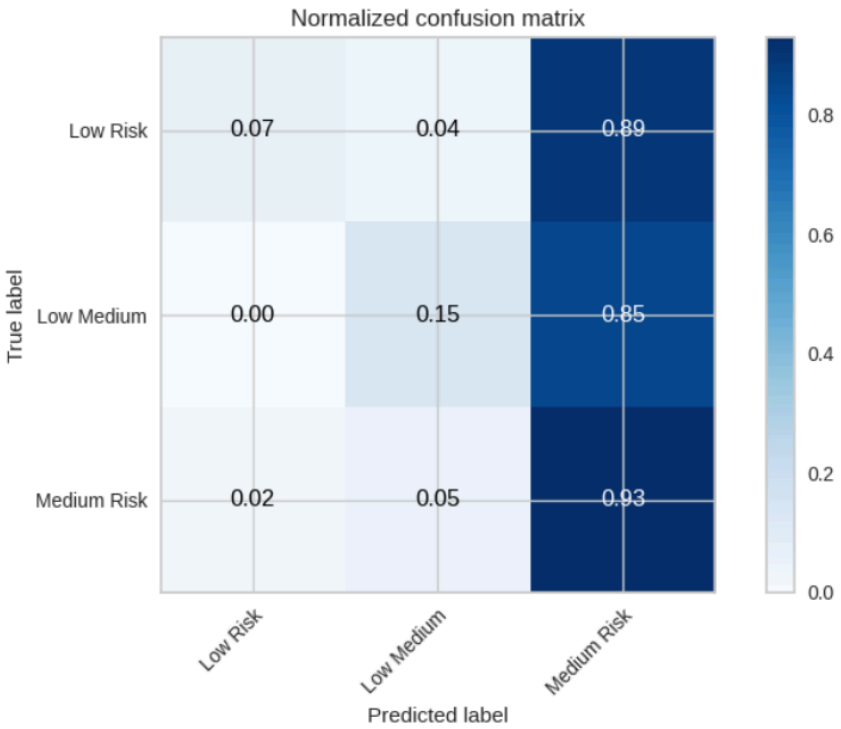

Pz_high_beta
Pz_gamma
T8_theta
AF4_high_beta
AF4_gamma
T7_low_beta
T8_alpha
T7_theta
T7_alpha
T8_high_beta
T8_low_beta
AF4_low_beta
T7_gamma
AF3_alpha
Pz_theta
T8_gamma
Pz_alpha
T7_high_beta
AF3_gamma
Pz_low_beta
0.0000
0.0025
0.0050
0.0075
0.0100
0.0125
0.0150
0.0175
mean(|SHAP value|) (average impact on model output magnitude)


AF4_theta <= 0.591
gini = 0.213
samples = 100.0%
value = [0.043, 0.074, 0.883]
class = Medium Risk
True
False
gini = 0.0
samples = 1.2%
value = [1.0, 0.0, 0.0]
class = Low Risk
Pz_gamma <= 0.303
gini = 0.195
samples = 98.8%
value = [0.032, 0.075, 0.893]
class = Medium Risk
T8_alpha <= 11.652
gini = 0.378
samples = 38.7%
value = [0.061, 0.172, 0.768]
class = Medium Risk
T8_low_beta <= 127.36
gini = 0.051
samples = 60.2%
value = [0.013, 0.013, 0.974]
class = Medium Risk
AF3_high_beta <= 0.865
gini = 0.223
samples = 25.4%
value = [0.046, 0.077, 0.877]
class = Medium Risk
AF4_theta <= 24.81
gini = 0.555
samples = 13.3%
value = [0.088, 0.353, 0.559]
class = Medium Risk
T7_theta <= 422.214
gini = 0.026
samples = 59.4%
value = [0.007, 0.007, 0.987]
class = Medium Risk
T7_high_beta <= 19.939
gini = 0.5
samples = 0.8%
value = [0.5, 0.5, 0.0]
class = Low Risk
(...)
(...)
(...)
(...)
(...)
(...)
(...)
(...)

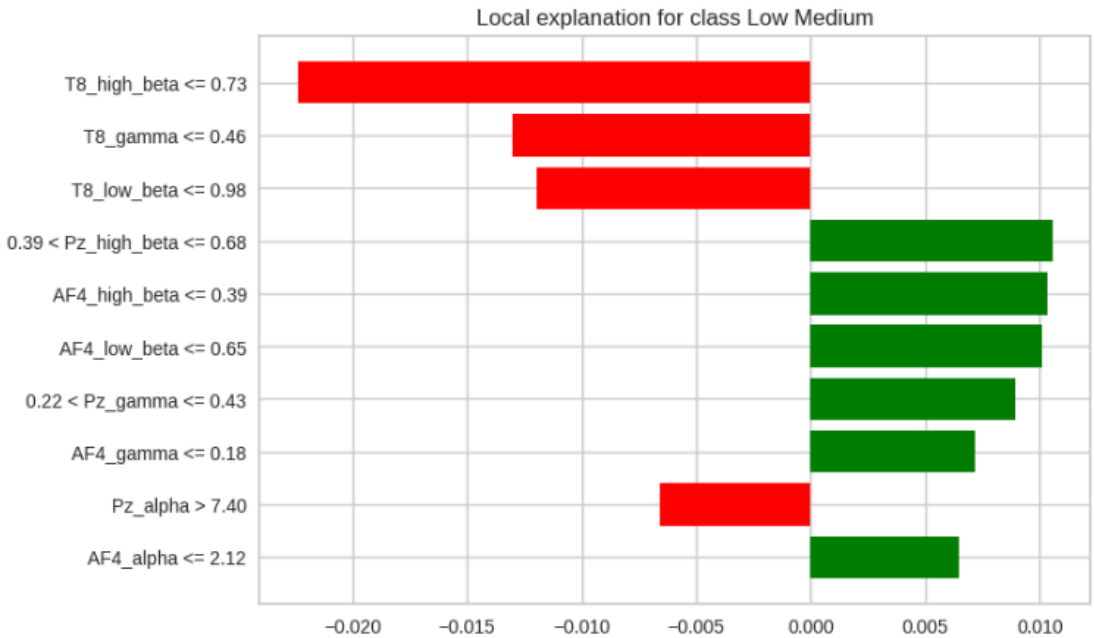
Poisoning rate 5% :

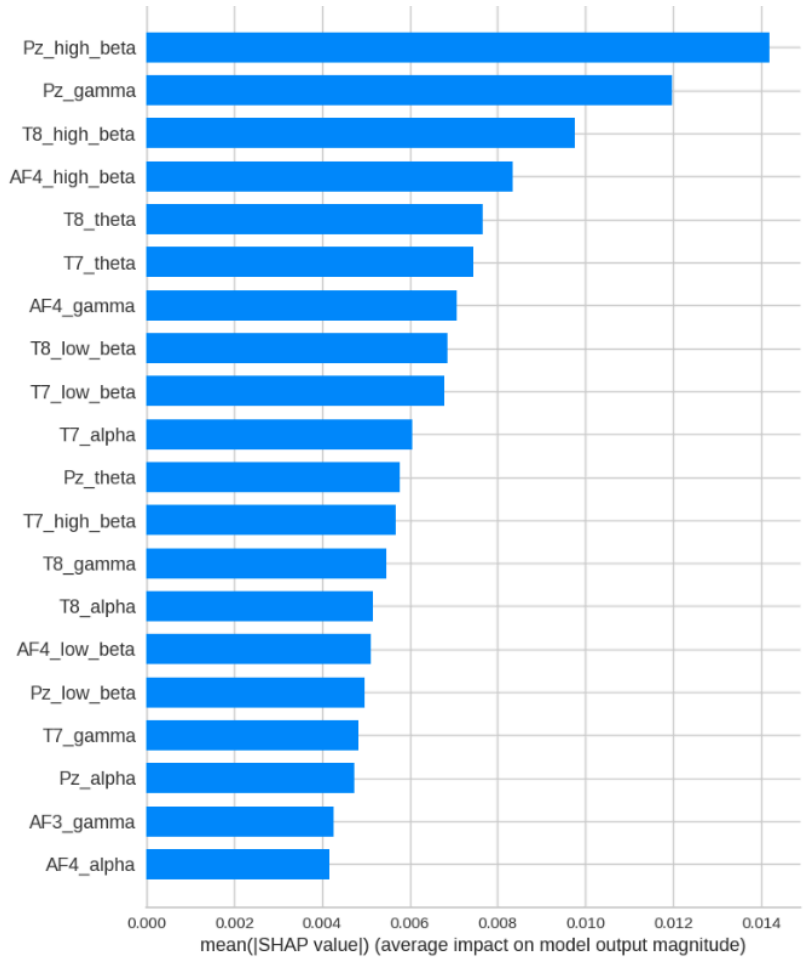


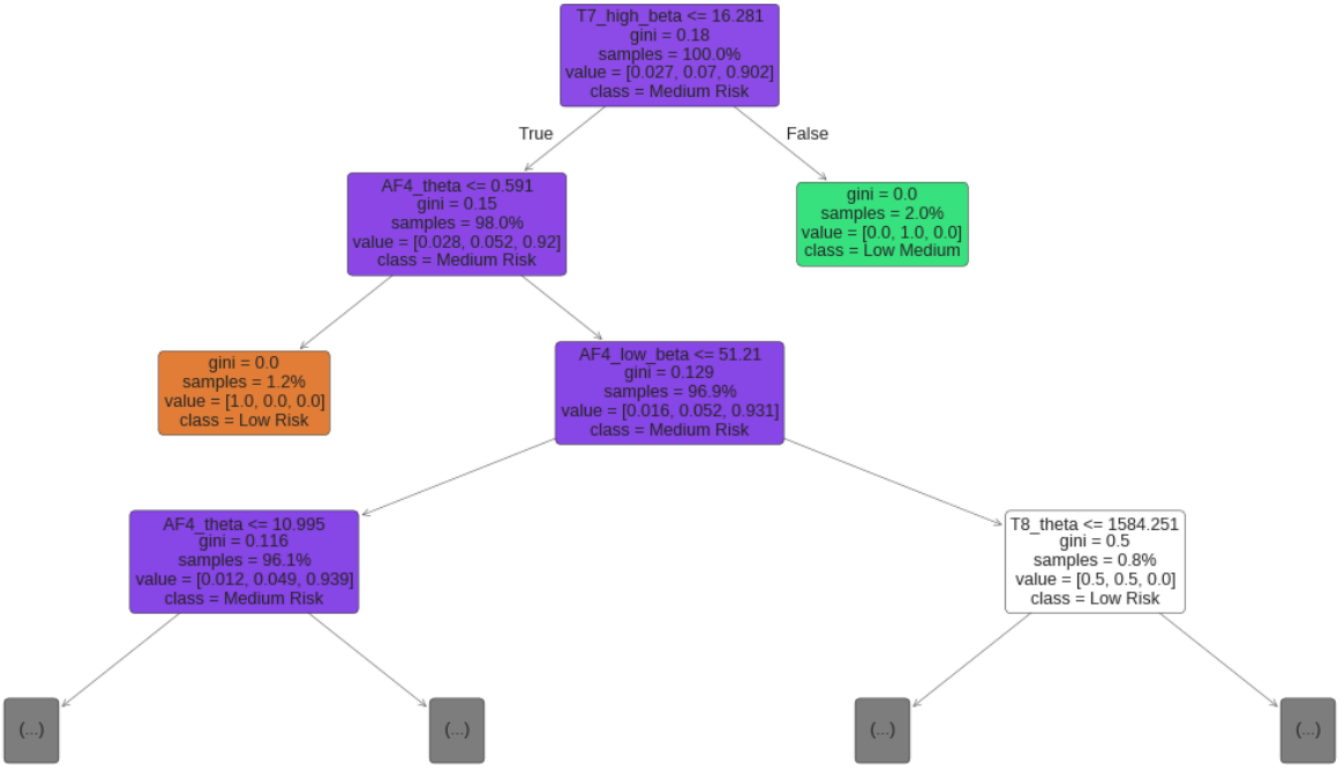




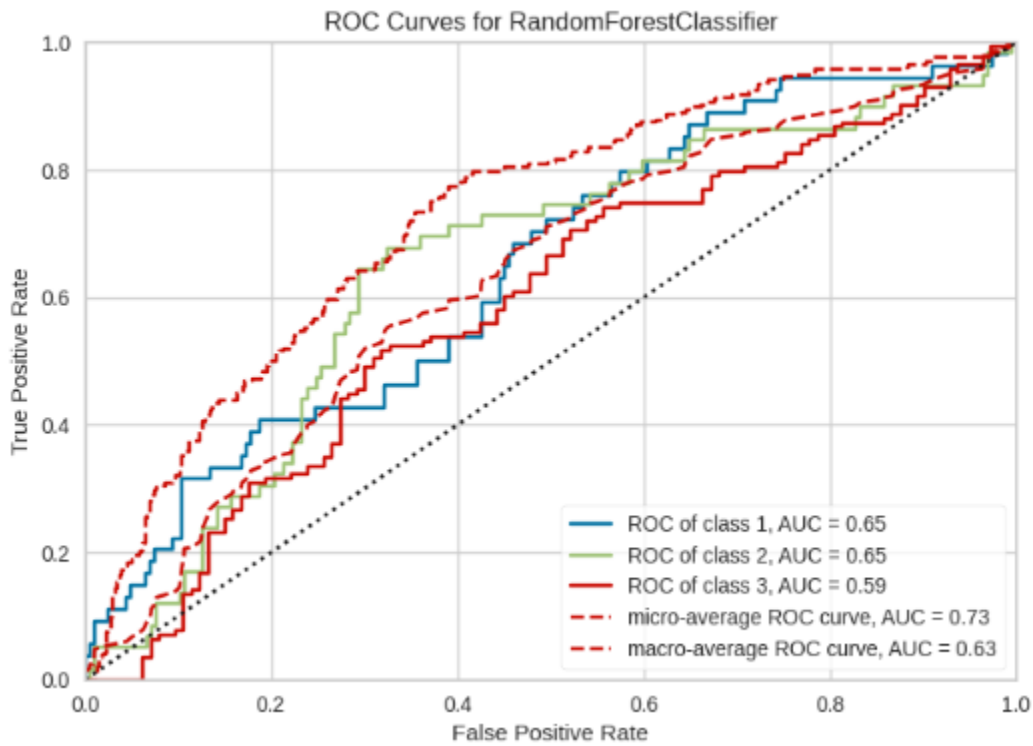
Weight	Feature
0.0117 ± 0.0070	Pz_low_beta
0.0102 ± 0.0063	T7_high_beta
0.0102 ± 0.0106	AF4_gamma
0.0094 ± 0.0117	T7_low_beta
0.0086 ± 0.0058	T7_theta
0.0063 ± 0.0106	AF4_alpha
0.0063 ± 0.0117	AF4_low_beta
0.0063 ± 0.0080	T7_gamma
0.0055 ± 0.0063	AF4_high_beta
0.0055 ± 0.0127	T7_alpha
0.0047 ± 0.0077	AF3_alpha
0.0047 ± 0.0058	AF3_high_beta
0.0047 ± 0.0077	Pz_gamma
0.0031 ± 0.0058	T8_gamma
0.0023 ± 0.0038	T8_alpha
0.0016 ± 0.0106	T8_low_beta
-0.0008 ± 0.0058	T8_high_beta
-0.0008 ± 0.0091	AF3_gamma
-0.0016 ± 0.0127	Pz_high_beta
-0.0031 ± 0.0091	T8_theta
... 4 more ...	

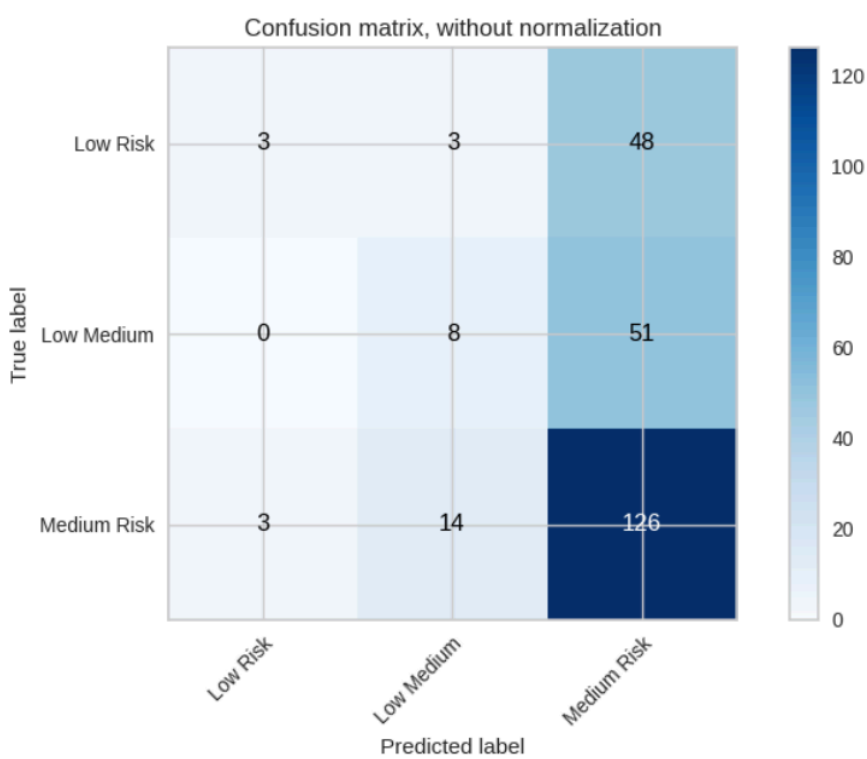

Local explanation for class Low Medium
T8_high_beta <= 0.73
T8_gamma <= 0.46
T8_low_beta <= 0.98
0.39 < Pz_high_beta <= 0.68
AF4_high_beta <= 0.39
AF4_low_beta <= 0.65
0.22 < Pz_gamma <= 0.43
AF4_gamma <= 0.18
Pz_alpha > 7.40
AF4_alpha <= 2.12
−0.020
−0.015
−0.010
−0.005
0.000
0.005
0.010

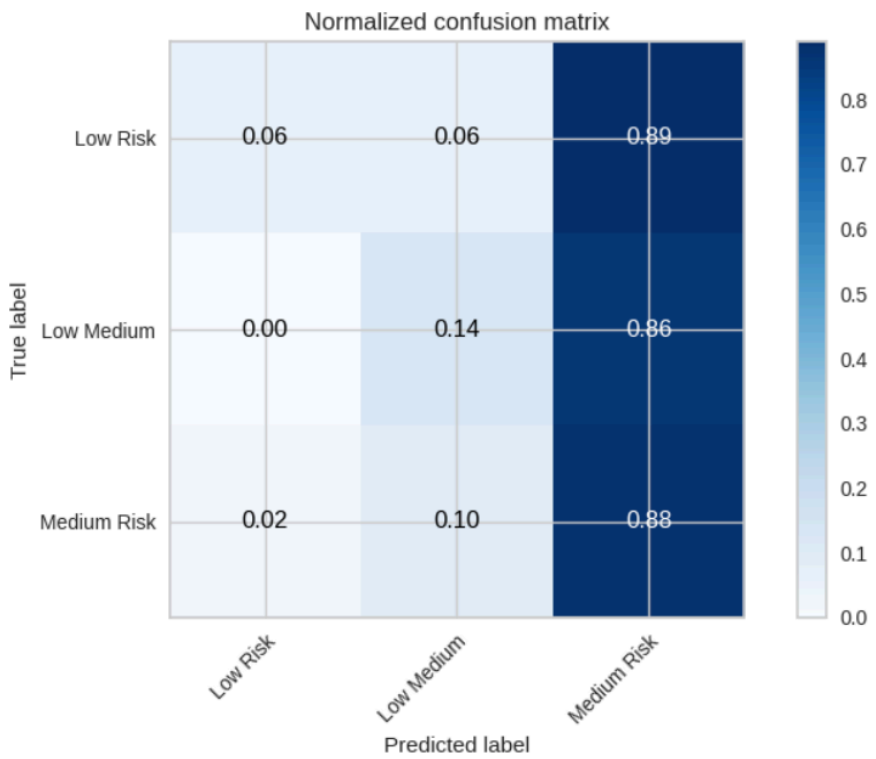

Pz_high_beta
Pz_gamma
T8_high_beta
AF4_high_beta
T8_theta
T7_theta
AF4_gamma
T8_low_beta
T7_low_beta
T7_alpha
Pz_theta
T7_high_beta
T8_gamma
T8_alpha
AF4_low_beta
Pz_low_beta
T7_gamma
Pz_alpha
AF3_gamma
AF4_alpha
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.014
mean(|SHAP value|) (average impact on model output magnitude)


T7_high_beta <= 16.281
gini = 0.18
samples = 100.0%
value = [0.027, 0.07, 0.902]
class = Medium Risk
True
False
AF4_theta <= 0.591
gini = 0.15
samples = 98.0%
value = [0.028, 0.052, 0.92]
class = Medium Risk
gini = 0.0
samples = 2.0%
value = [0.0, 1.0, 0.0]
class = Low Medium
gini = 0.0
samples = 1.2%
value = [1.0, 0.0, 0.0]
class = Low Risk
AF4_low_beta <= 51.21
gini = 0.129
samples = 96.9%
value = [0.016, 0.052, 0.931]
class = Medium Risk
AF4_theta <= 10.995
gini = 0.116
samples = 96.1%
value = [0.012, 0.049, 0.939]
class = Medium Risk
T8_theta <= 1584.251
gini = 0.5
samples = 0.8%
value = [0.5, 0.5, 0.0]
class = Low Risk
(...)
(...)
(...)
(...)

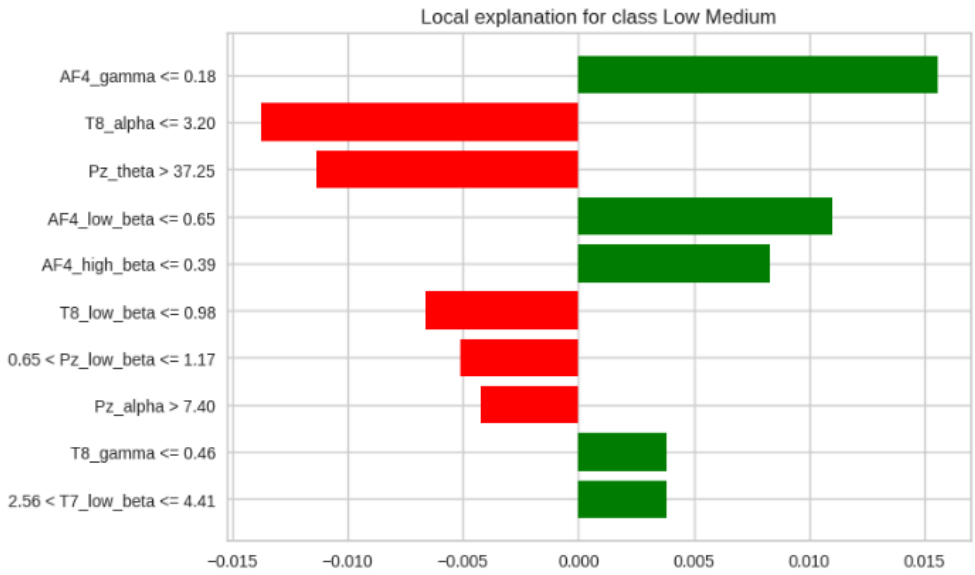
Poisoning rate 25% :

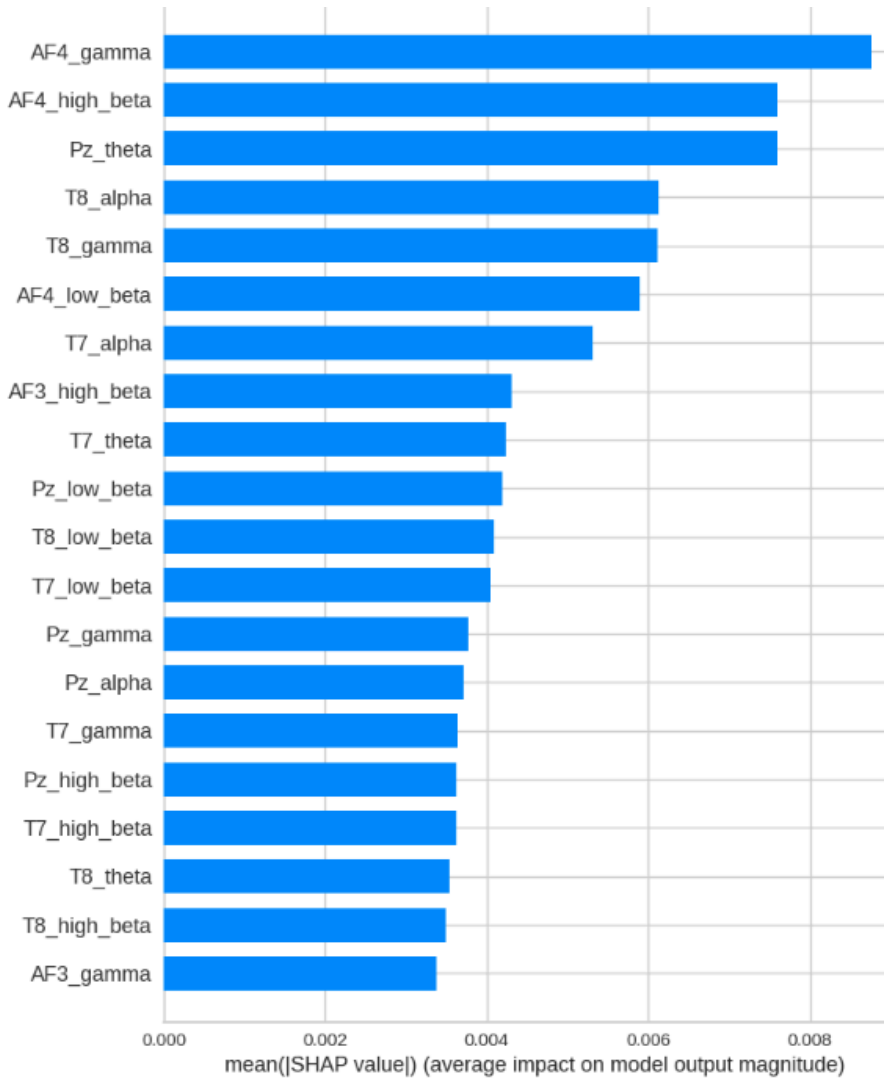


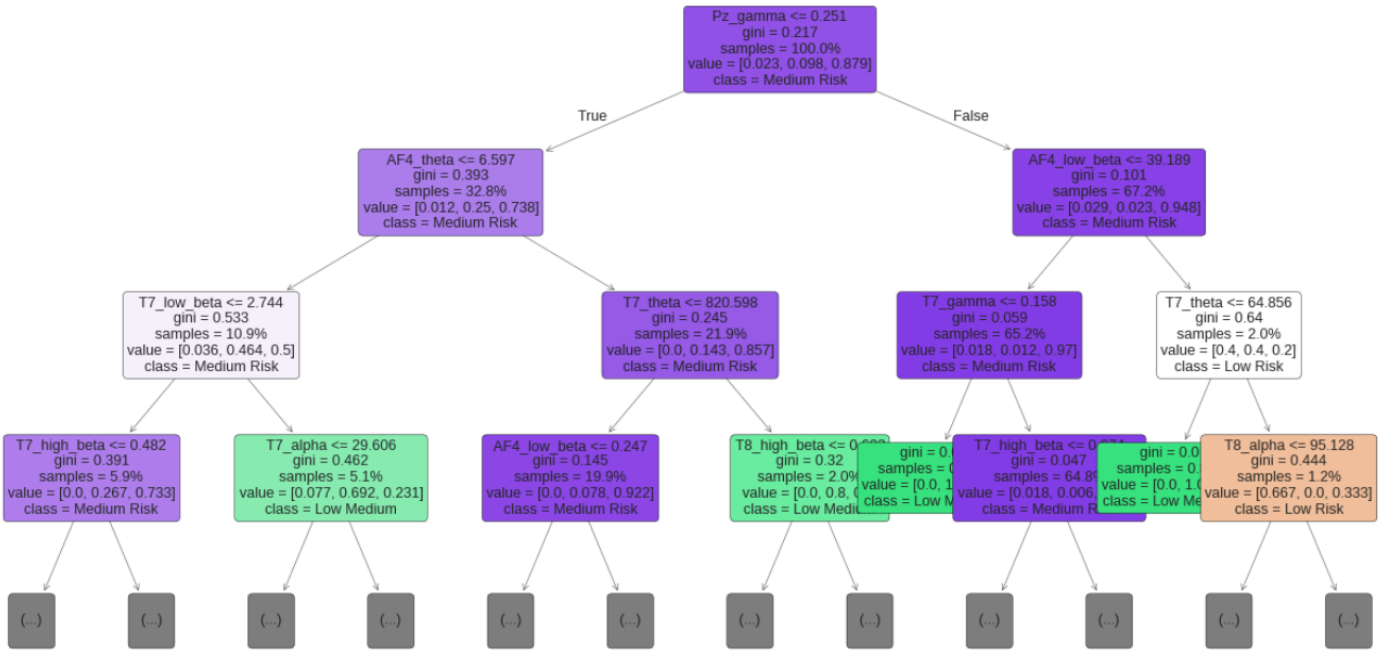




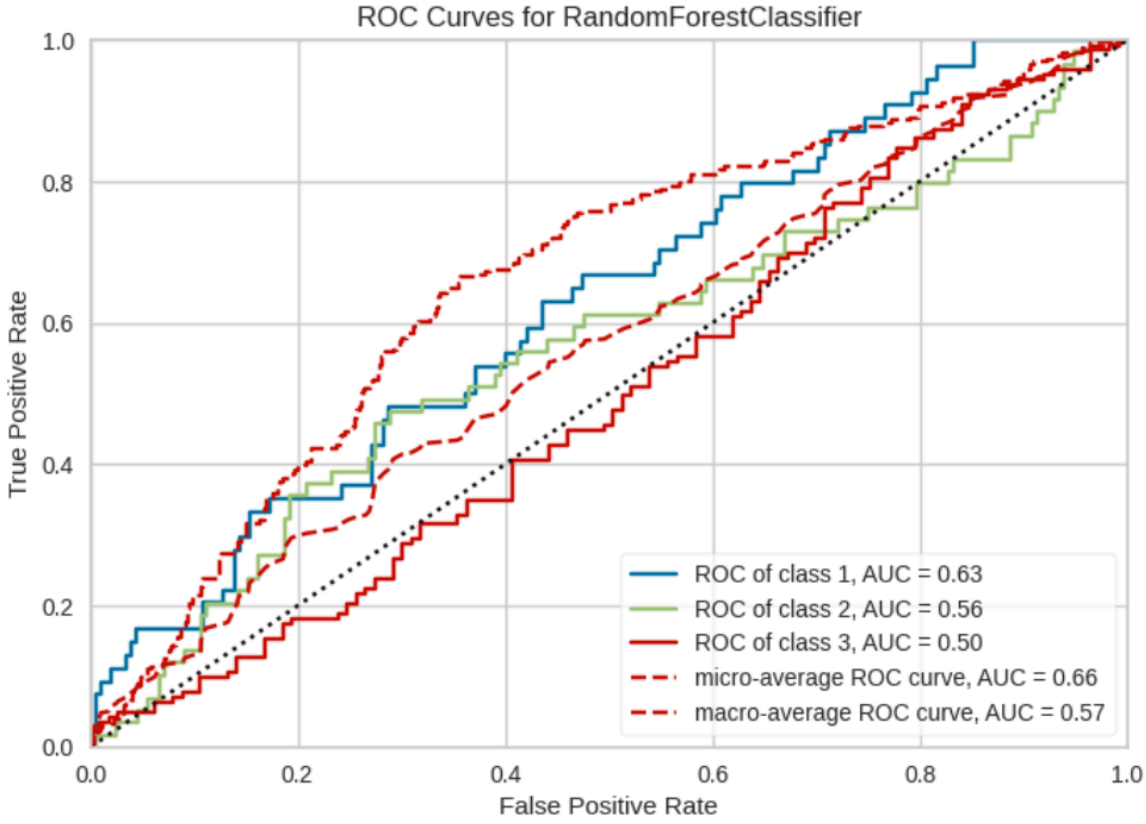
Weight	Feature
0.0125 ± 0.0091	Pz_gamma
0.0086 ± 0.0115	AF3_low_beta
0.0078 ± 0.0110	T7_low_beta
0.0063 ± 0.0094	Pz_low_beta
0.0063 ± 0.0127	AF4_low_beta
0.0055 ± 0.0063	AF4_theta
0.0023 ± 0.0094	T8_theta
0.0008 ± 0.0125	T7_high_beta
0.0000 ± 0.0086	AF3_alpha
-0.0008 ± 0.0058	Pz_high_beta
-0.0016 ± 0.0189	T7_theta
-0.0016 ± 0.0117	Pz_theta
-0.0023 ± 0.0063	AF4_alpha
-0.0023 ± 0.0063	Pz_alpha
-0.0023 ± 0.0080	AF3_gamma
-0.0023 ± 0.0127	AF3_high_beta
-0.0031 ± 0.0058	T7_alpha
-0.0039 ± 0.0156	T8_high_beta
-0.0047 ± 0.0091	T7_gamma
-0.0063 ± 0.0106	T8_low_beta
... 4 more ...	

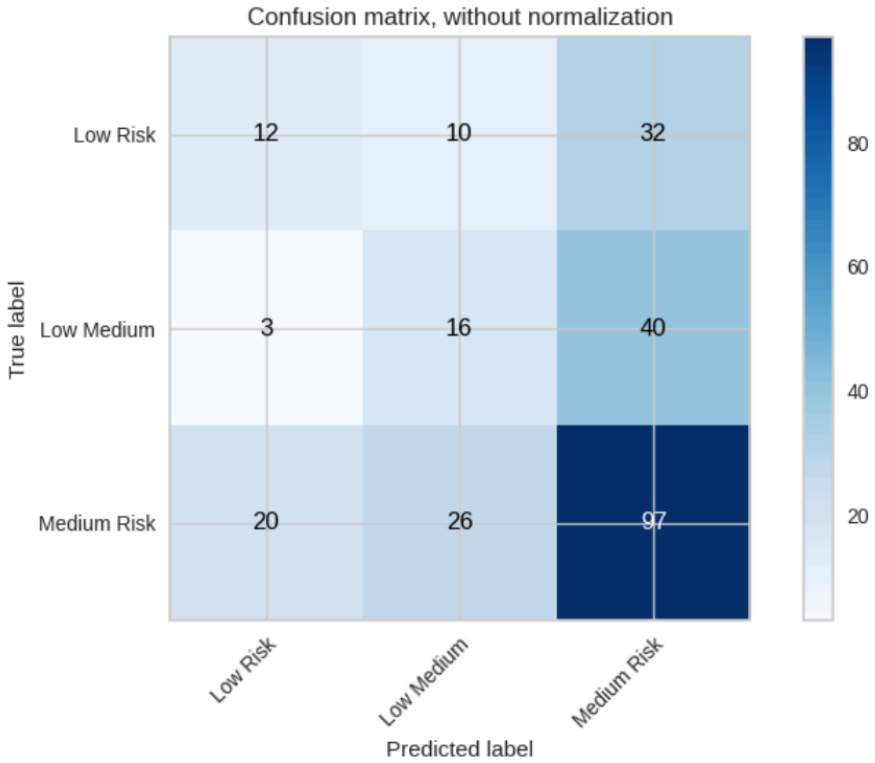

Local explanation for class Low Medium
AF4_gamma <= 0.18
T8_alpha <= 3.20
Pz_theta > 37.25
AF4_low_beta <= 0.65
AF4_high_beta <= 0.39
T8_low_beta <= 0.98
0.65 < Pz_low_beta <= 1.17
Pz_alpha > 7.40
T8_gamma <= 0.46
2.56 < T7_low_beta <= 4.41
−0.015
−0.010
−0.005
0.000
0.005
0.010
0.015

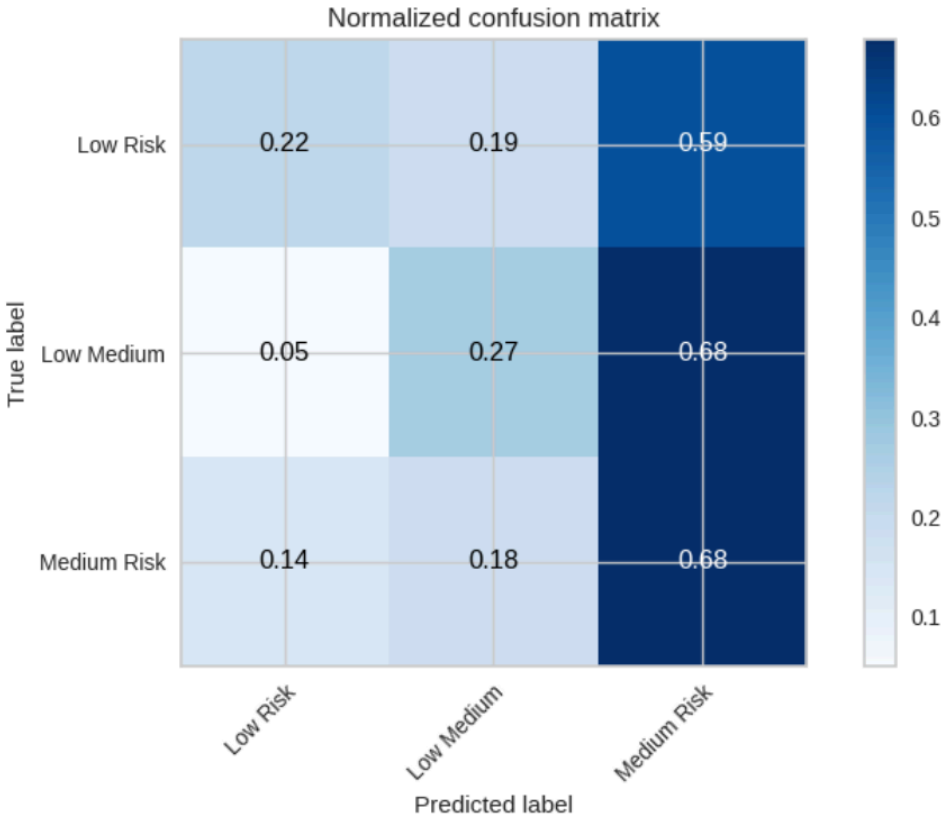

AF4_gamma
AF4_high_beta
Pz_theta
T8_alpha
T8_gamma
AF4_low_beta
T7_alpha
AF3_high_beta
T7_theta
Pz_low_beta
T8_low_beta
T7_low_beta
Pz_gamma
Pz_alpha
T7_gamma
Pz_high_beta
T7_high_beta
T8_theta
T8_high_beta
AF3_gamma
0.000
0.002
0.004
0.006
0.008
mean(|SHAP value|) (average impact on model output magnitude)


Pz_gamma <= 0.251
gini = 0.217
samples = 100.0%
value = [0.023, 0.098, 0.879]
class = Medium Risk
True
False
AF4_theta <= 6.597
gini = 0.393
samples = 32.8%
value = [0.012, 0.25, 0.738]
class = Medium Risk
AF4_low_beta <= 39.189
gini = 0.101
samples = 67.2%
value = [0.029, 0.023, 0.948]
class = Medium Risk
T7_low_beta <= 2.744
gini = 0.533
samples = 10.9%
value = [0.036, 0.464, 0.5]
class = Medium Risk
T7_theta <= 820.598
gini = 0.245
samples = 21.9%
value = [0.0, 0.143, 0.857]
class = Medium Risk
T7_gamma <= 0.158
gini = 0.059
samples = 65.2%
value = [0.018, 0.012, 0.97]
class = Medium Risk
T7_theta <= 64.856
gini = 0.64
samples = 2.0%
value = [0.4, 0.4, 0.2]
class = Low Risk
T7_high_beta <= 0.482
gini = 0.391
samples = 5.9%
value = [0.0, 0.267, 0.733]
class = Medium Risk
T7_alpha <= 29.606
gini = 0.462
samples = 5.1%
value = [0.077, 0.692, 0.231]
class = Low Medium
AF4_low_beta <= 0.247
gini = 0.145
samples = 19.9%
value = [0.0, 0.078, 0.922]
class = Medium Risk
T8_high_beta <= 0.
gini = 0.32
samples = 2.0%
value = [0.0, 0.8,
class = Low Medi
gini = 0.
samples =
value = [0.0, 1
class = Low M
T7_high_beta <= 0.
gini = 0.047
samples = 64.8
value = [0.018, 0.006,
class = Medium R
gini = 0.0
samples = 0.
value = [0.0, 1.
class = Low Me
T8_alpha <= 95.128
gini = 0.444
samples = 1.2%
value = [0.667, 0.0, 0.333]
class = Low Risk
(...)
(...)
(...)
(...)
(...)
(...)
(...)
(...)
(...)
(...)
(...)
(...)

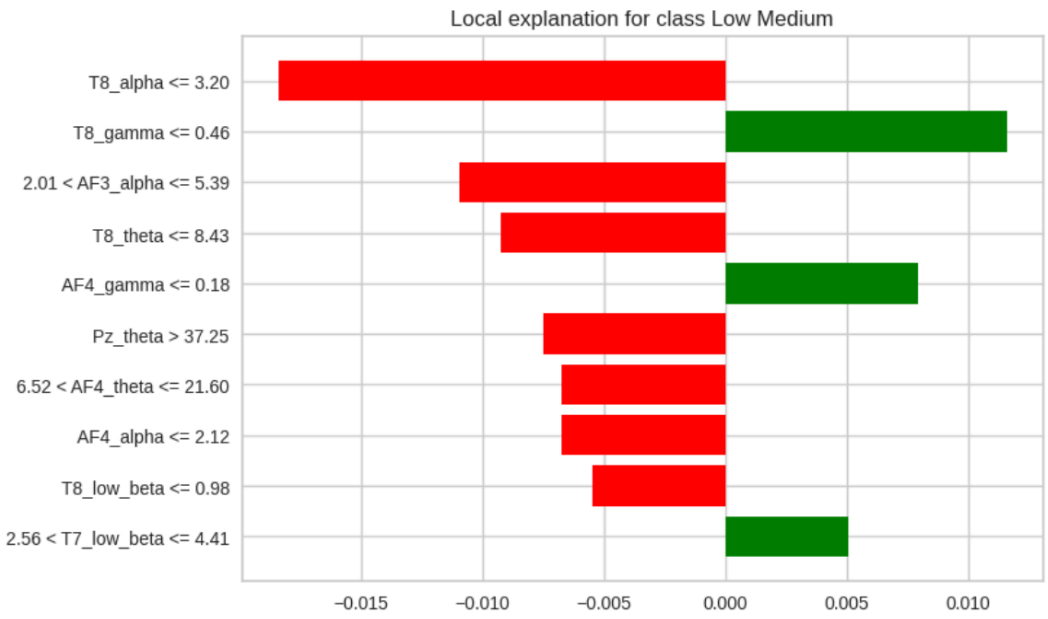
Poisoning rate 50% :

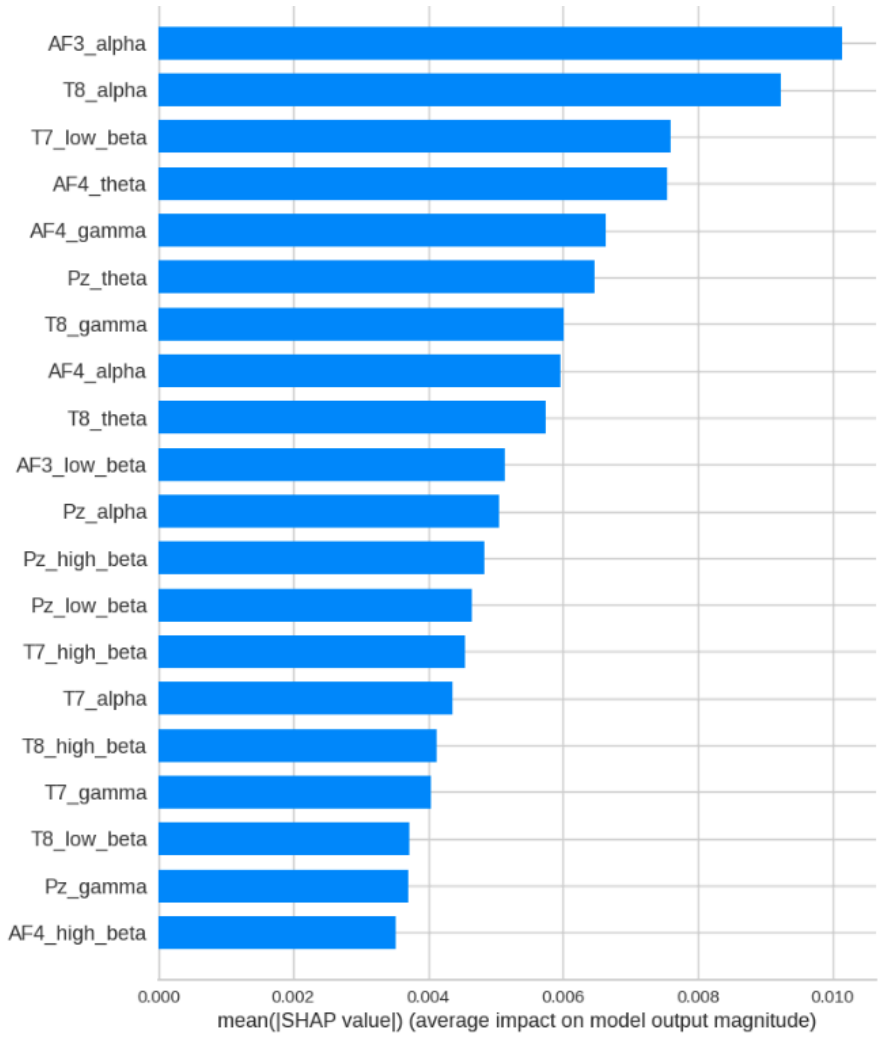


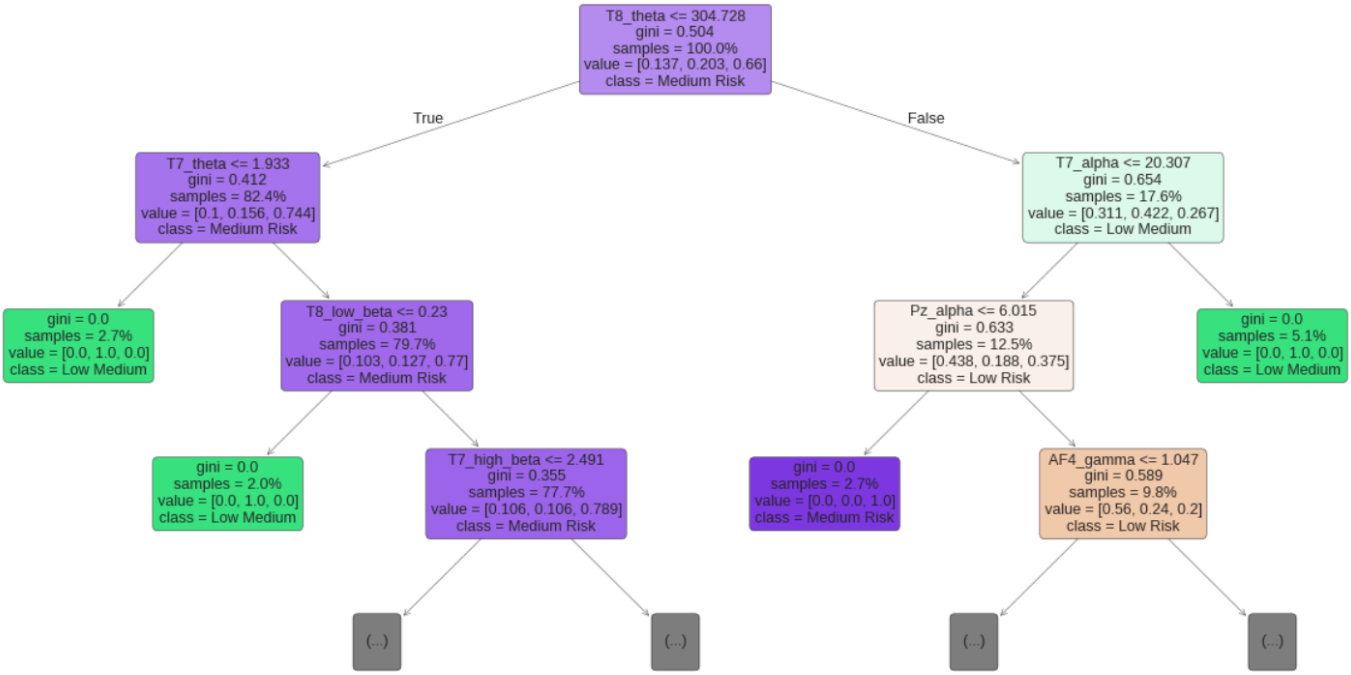




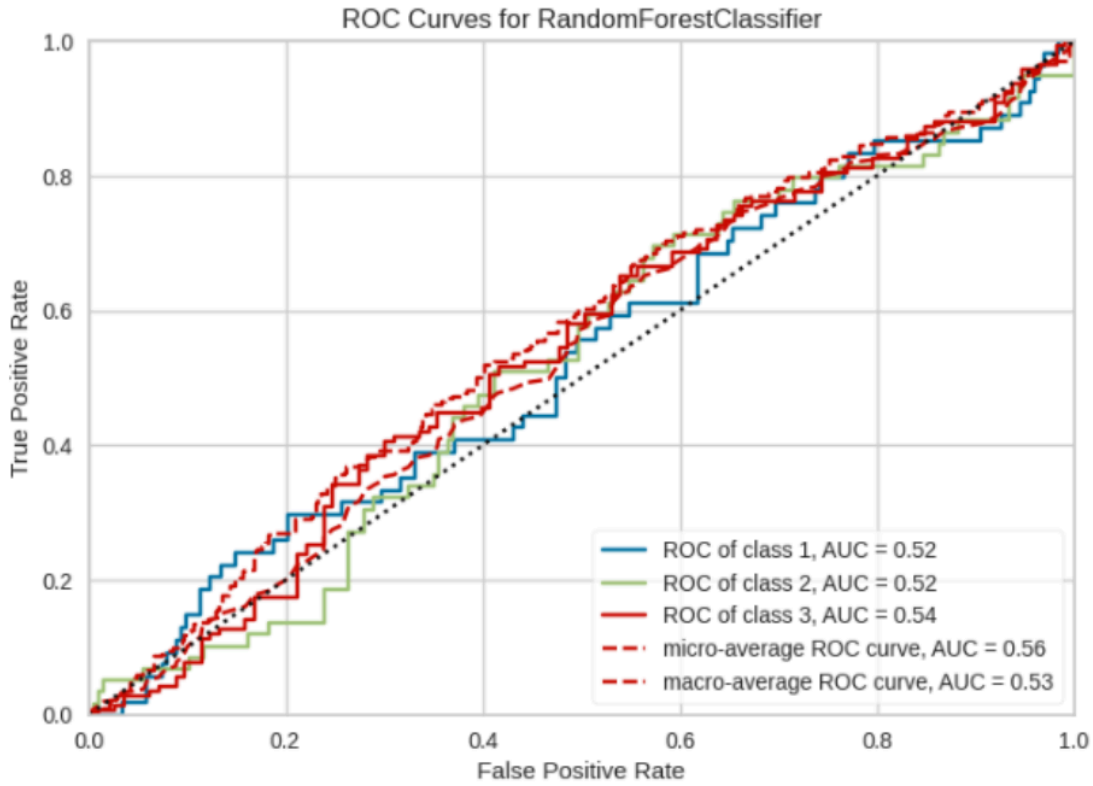
Weight	Feature
0.0094 ± 0.0161	AF3_high_beta
0.0086 ± 0.0159	Pz_low_beta
0.0086 ± 0.0031	T7_theta
0.0047 ± 0.0174	Pz_alpha
0.0008 ± 0.0058	AF4_low_beta
0.0000 ± 0.0271	T7_high_beta
-0.0008 ± 0.0159	AF3_low_beta
-0.0008 ± 0.0326	Pz_theta
-0.0008 ± 0.0125	AF3_gamma
-0.0016 ± 0.0153	AF4_alpha
-0.0023 ± 0.0080	Pz_gamma
-0.0023 ± 0.0153	AF4_gamma
-0.0031 ± 0.0115	T7_alpha
-0.0047 ± 0.0091	Pz_high_beta
-0.0047 ± 0.0091	T8_high_beta
-0.0055 ± 0.0161	T7_gamma
-0.0070 ± 0.0091	AF4_high_beta
-0.0070 ± 0.0134	T8_alpha
-0.0078 ± 0.0110	AF3_alpha
-0.0086 ± 0.0134	T8_low_beta
... 4 more ...	

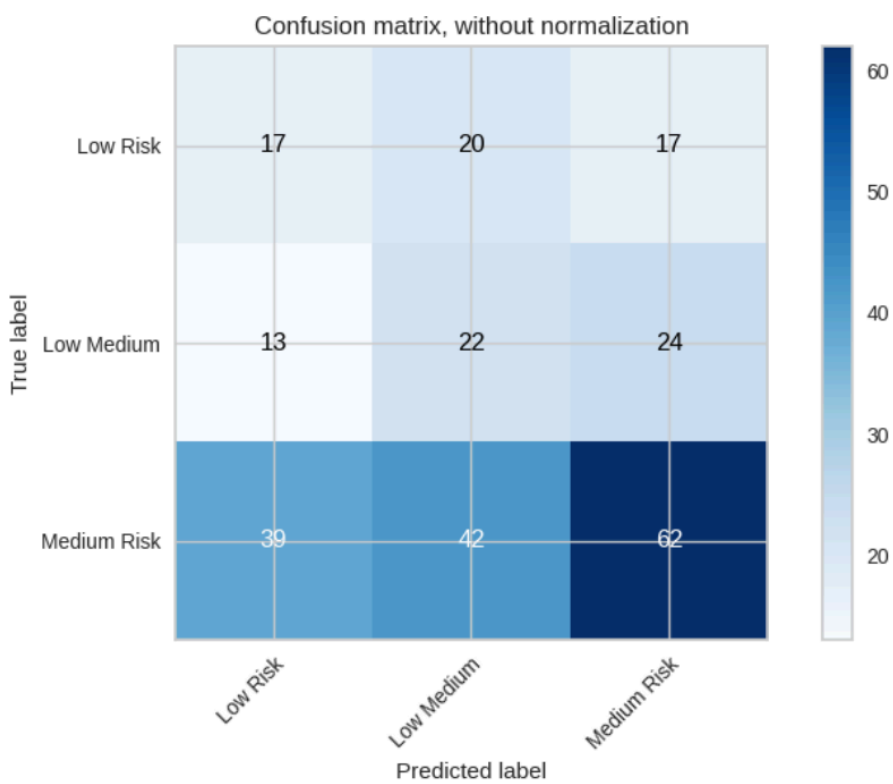

Local explanation for class Low Medium
T8_alpha <= 3.20
T8_gamma <= 0.46
2.01 < AF3_alpha <= 5.39
T8_theta <= 8.43
AF4_gamma <= 0.18
Pz_theta > 37.25
6.52 < AF4_theta <= 21.60
AF4_alpha <= 2.12
T8_low_beta <= 0.98
2.56 < T7_low_beta <= 4.41
−0.015
−0.010
−0.005
0.000
0.005
0.010

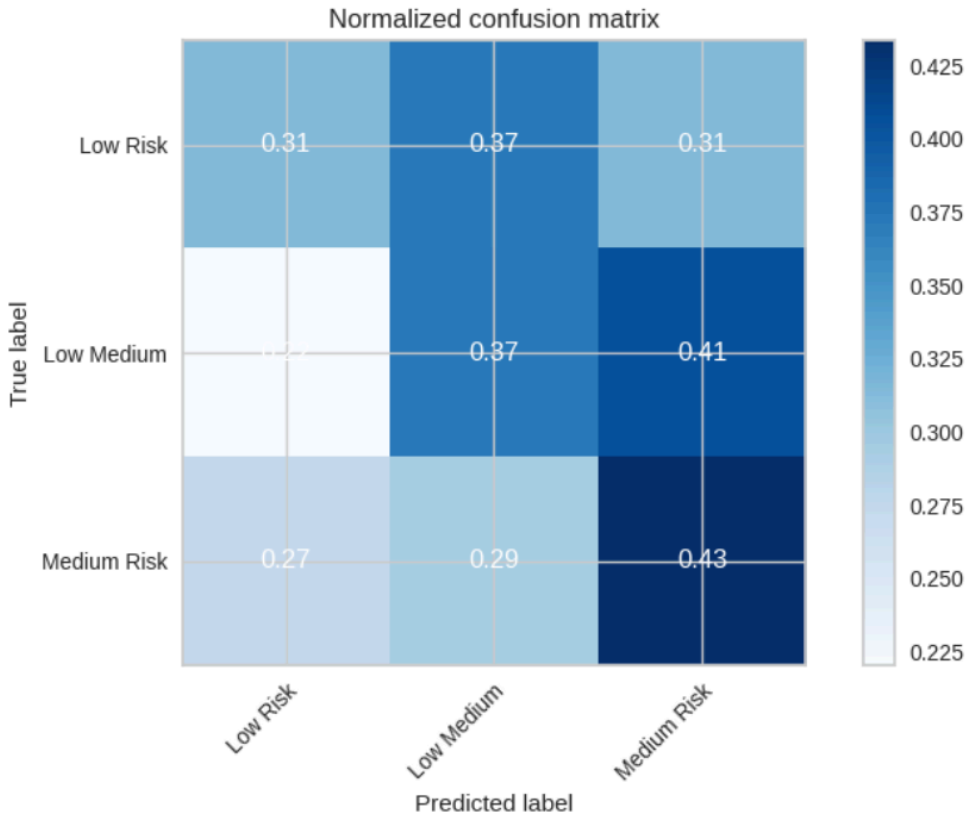

AF3_alpha
T8_alpha
T7_low_beta
AF4_theta
AF4_gamma
Pz_theta
T8_gamma
AF4_alpha
T8_theta
AF3_low_beta
Pz_alpha
Pz_high_beta
Pz_low_beta
T7_high_beta
T7_alpha
T8_high_beta
T7_gamma
T8_low_beta
Pz_gamma
AF4_high_beta
0.000
0.002
0.004
0.006
0.008
0.010
mean(|SHAP value|) (average impact on model output magnitude)


T8_theta <= 304.728
gini = 0.504
samples = 100.0%
value = [0.137, 0.203, 0.66]
class = Medium Risk
True
False
T7_theta <= 1.933
gini = 0.412
samples = 82.4%
value = [0.1, 0.156, 0.744]
class = Medium Risk
T7_alpha <= 20.307
gini = 0.654
samples = 17.6%
value = [0.311, 0.422, 0.267]
class = Low Medium
gini = 0.0
samples = 2.7%
value = [0.0, 1.0, 0.0]
class = Low Medium
T8_low_beta <= 0.23
gini = 0.381
samples = 79.7%
value = [0.103, 0.127, 0.77]
class = Medium Risk
Pz_alpha <= 6.015
gini = 0.633
samples = 12.5%
value = [0.438, 0.188, 0.375]
class = Low Risk
gini = 0.0
samples = 5.1%
value = [0.0, 1.0, 0.0]
class = Low Medium
gini = 0.0
samples = 2.0%
value = [0.0, 1.0, 0.0]
class = Low Medium
T7_high_beta <= 2.491
gini = 0.355
samples = 77.7%
value = [0.106, 0.106, 0.789]
class = Medium Risk
gini = 0.0
samples = 2.7%
value = [0.0, 0.0, 1.0]
class = Medium Risk
AF4_gamma <= 1.047
gini = 0.589
samples = 9.8%
value = [0.56, 0.24, 0.2]
class = Low Risk
(...)
(...)
(...)
(...)

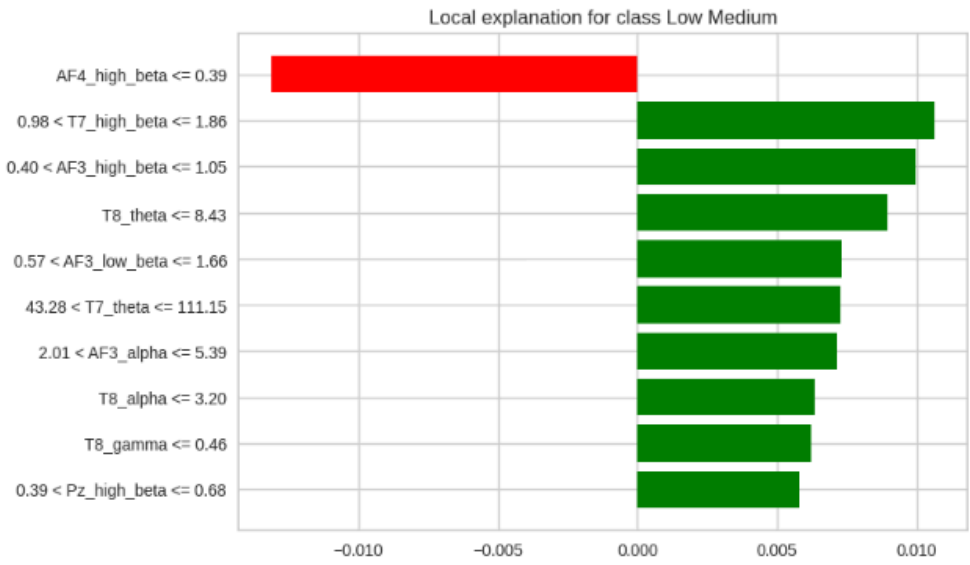
Poisoning rate 75% :

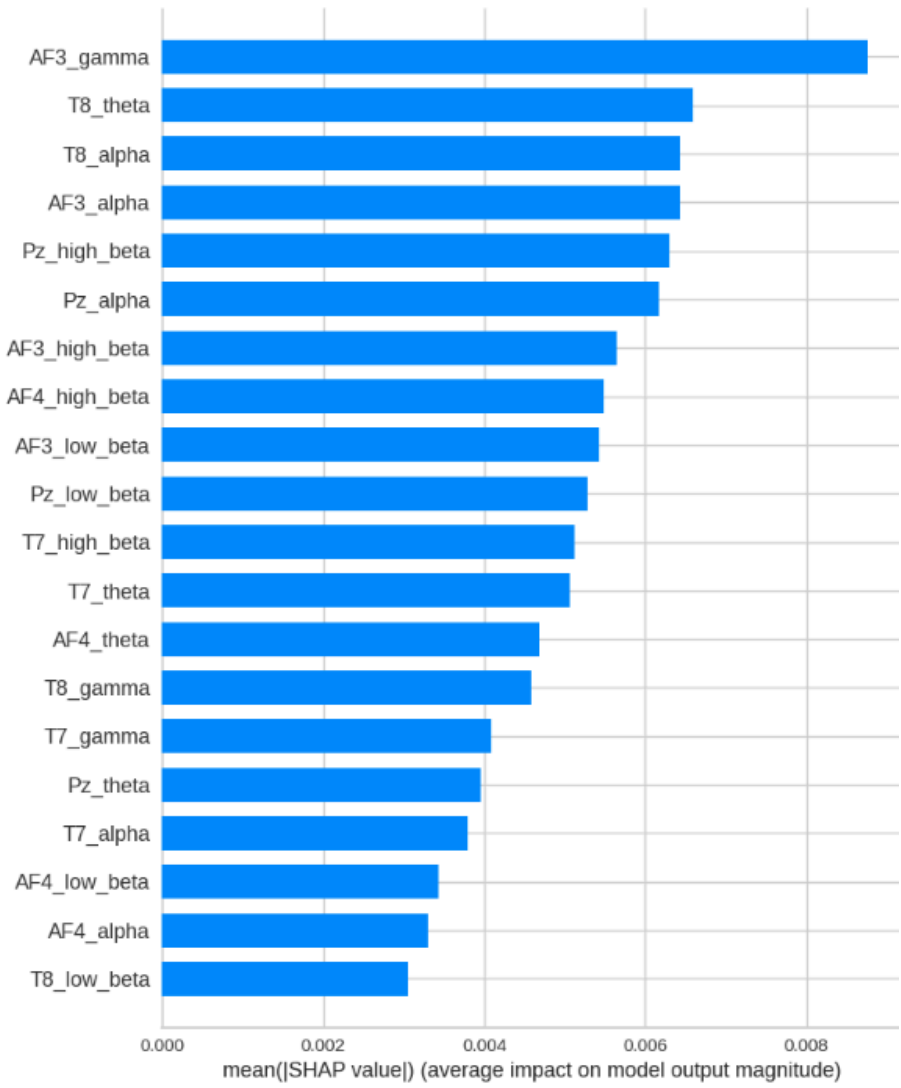


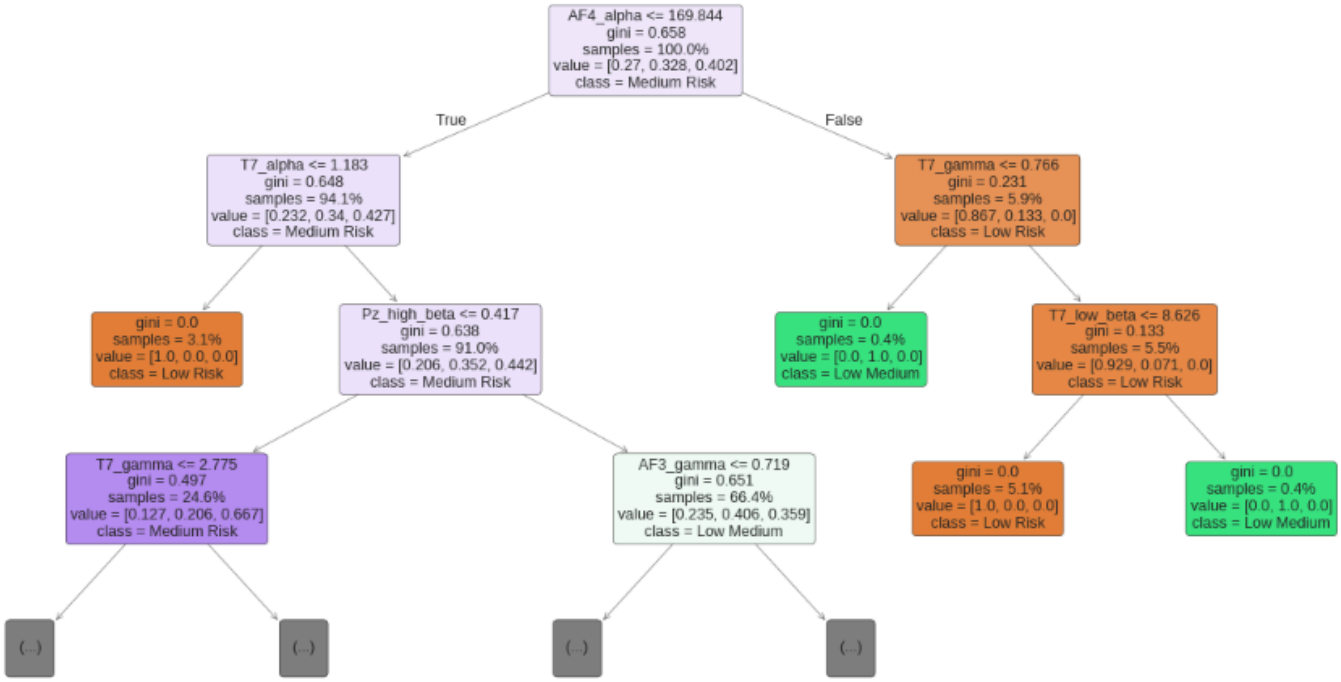




Weight	Feature
0.0156 ± 0.0110	AF3_gamma
0.0125 ± 0.0181	AF4_low_beta
0.0102 ± 0.0224	AF4_alpha
0.0102 ± 0.0235	T8_high_beta
0.0094 ± 0.0161	AF4_theta
0.0078 ± 0.0164	T7_high_beta
0.0063 ± 0.0145	Pz_alpha
0.0055 ± 0.0136	Pz_high_beta
0.0055 ± 0.0127	AF3_alpha
0.0047 ± 0.0200	T8_alpha
0.0008 ± 0.0143	T8_low_beta
-0.0008 ± 0.0125	T8_theta
-0.0016 ± 0.0136	T8_gamma
-0.0023 ± 0.0080	Pz_gamma
-0.0031 ± 0.0115	T7_gamma
-0.0031 ± 0.0229	Pz_low_beta
-0.0031 ± 0.0254	T7_theta
-0.0063 ± 0.0153	T7_low_beta
-0.0070 ± 0.0340	AF3_low_beta
-0.0094 ± 0.0250	Pz_theta
... 4 more ...	


Local explanation for class Low Medium
AF4_high_beta <= 0.39
0.98 < T7_high_beta <= 1.86
0.40 < AF3_high_beta <= 1.05
T8_theta <= 8.43
0.57 < AF3_low_beta <= 1.66
43.28 < T7_theta <= 111.15
2.01 < AF3_alpha <= 5.39
T8_alpha <= 3.20
T8_gamma <= 0.46
0.39 < Pz_high_beta <= 0.68
−0.010
−0.005
0.000
0.005
0.010


AF3_gamma
T8_theta
T8_alpha
AF3_alpha
Pz_high_beta
Pz_alpha
AF3_high_beta
AF4_high_beta
AF3_low_beta
Pz_low_beta
T7_high_beta
T7_theta
AF4_theta
T8_gamma
T7_gamma
Pz_theta
T7_alpha
AF4_low_beta
AF4_alpha
T8_low_beta
0.000
0.002
0.004
0.006
0.008
mean(|SHAP value|) (average impact on model output magnitude)


AF4_alpha <= 169.844
gini = 0.658
samples = 100.0%
value = [0.27, 0.328, 0.402]
class = Medium Risk
True
False
T7_alpha <= 1.183
gini = 0.648
samples = 94.1%
value = [0.232, 0.34, 0.427]
class = Medium Risk
T7_gamma <= 0.766
gini = 0.231
samples = 5.9%
value = [0.867, 0.133, 0.0]
class = Low Risk
gini = 0.0
samples = 3.1%
value = [1.0, 0.0, 0.0]
class = Low Risk
Pz_high_beta <= 0.417
gini = 0.638
samples = 91.0%
value = [0.206, 0.352, 0.442]
class = Medium Risk
gini = 0.0
samples = 0.4%
value = [0.0, 1.0, 0.0]
class = Low Medium
T7_low_beta <= 8.626
gini = 0.133
samples = 5.5%
value = [0.929, 0.071, 0.0]
class = Low Risk
T7_gamma <= 2.775
gini = 0.497
samples = 24.6%
value = [0.127, 0.206, 0.667]
class = Medium Risk
AF3_gamma <= 0.719
gini = 0.651
samples = 66.4%
value = [0.235, 0.406, 0.359]
class = Low Medium
gini = 0.0
samples = 5.1%
value = [1.0, 0.0, 0.0]
class = Low Risk
gini = 0.0
samples = 0.4%
value = [0.0, 1.0, 0.0]
class = Low Medium
(...)
(...)
(...)
(...)